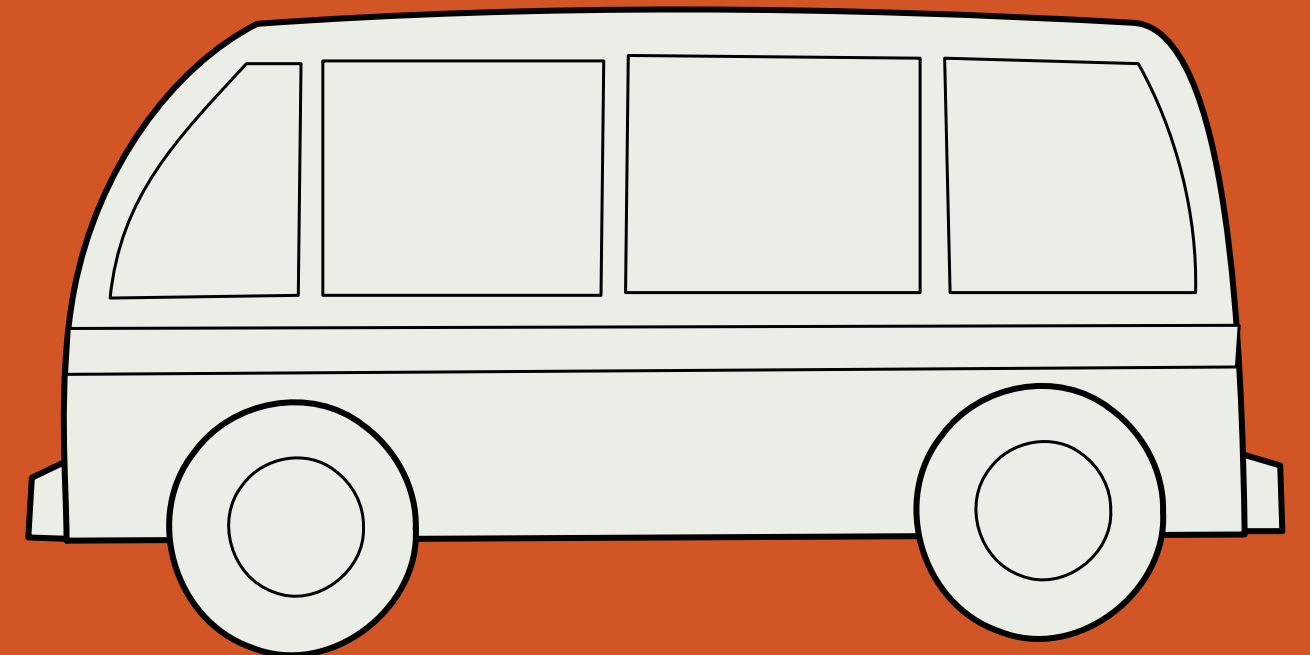




TRACKING AND LOCATING SYSTEM ON WEBSITE



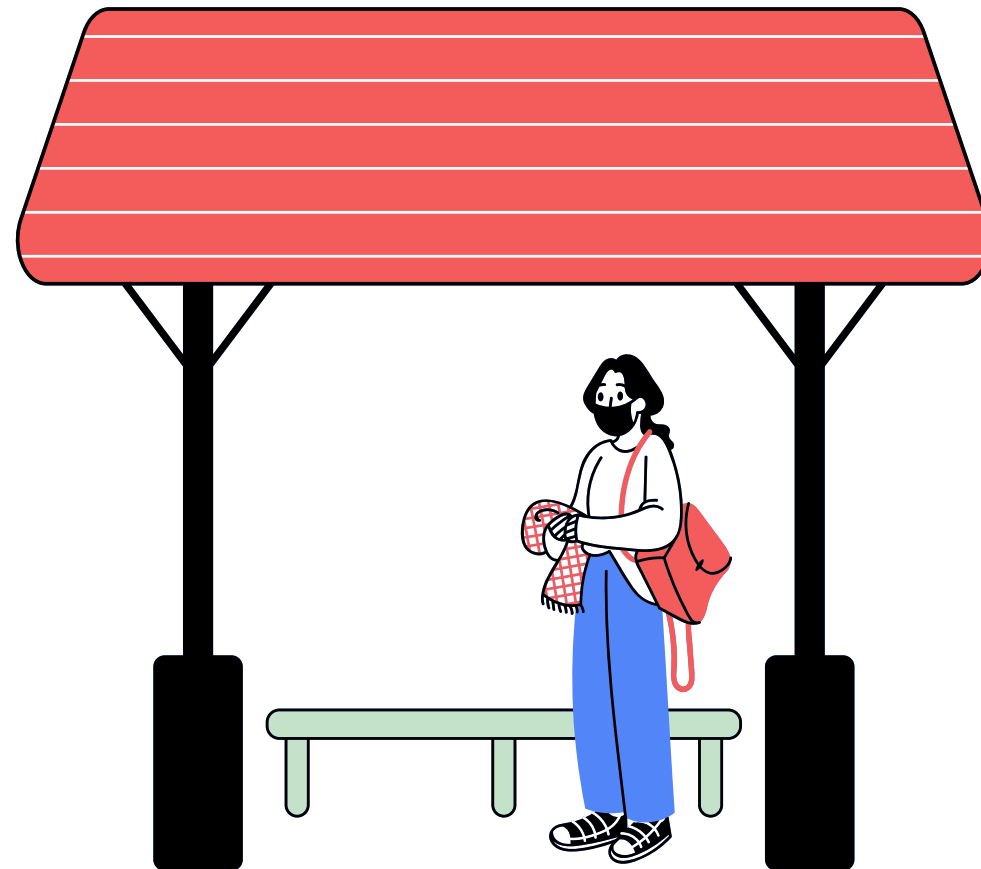
Objective

To develop tracking and locating systems to be applied in the organization.

Scope

A vehicle tracking device to indicate the location of the vehicle will be displayed on the website.

The system can track and locate devices more than 1 device



Resources

Equipment



Hardware

1. NodeMCU/ESP8266
2. GPS Module
3. WIFI access point

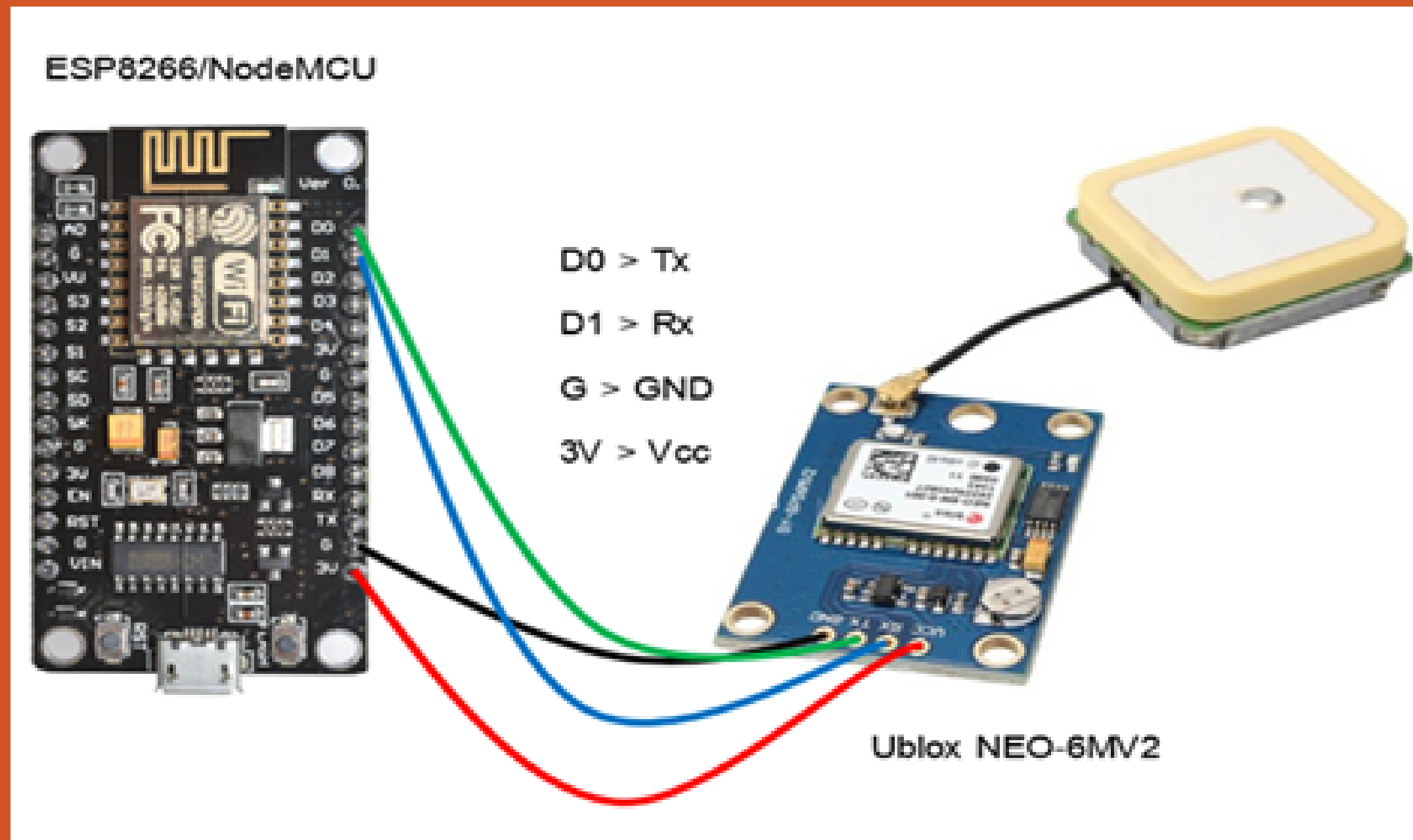
Software

1. Visual Studio Code
2. Arduino IDE
3. NETPIE2020
4. Longdo map



Circuit Diagram

GPS Module NEO6MV2
with ESP8266



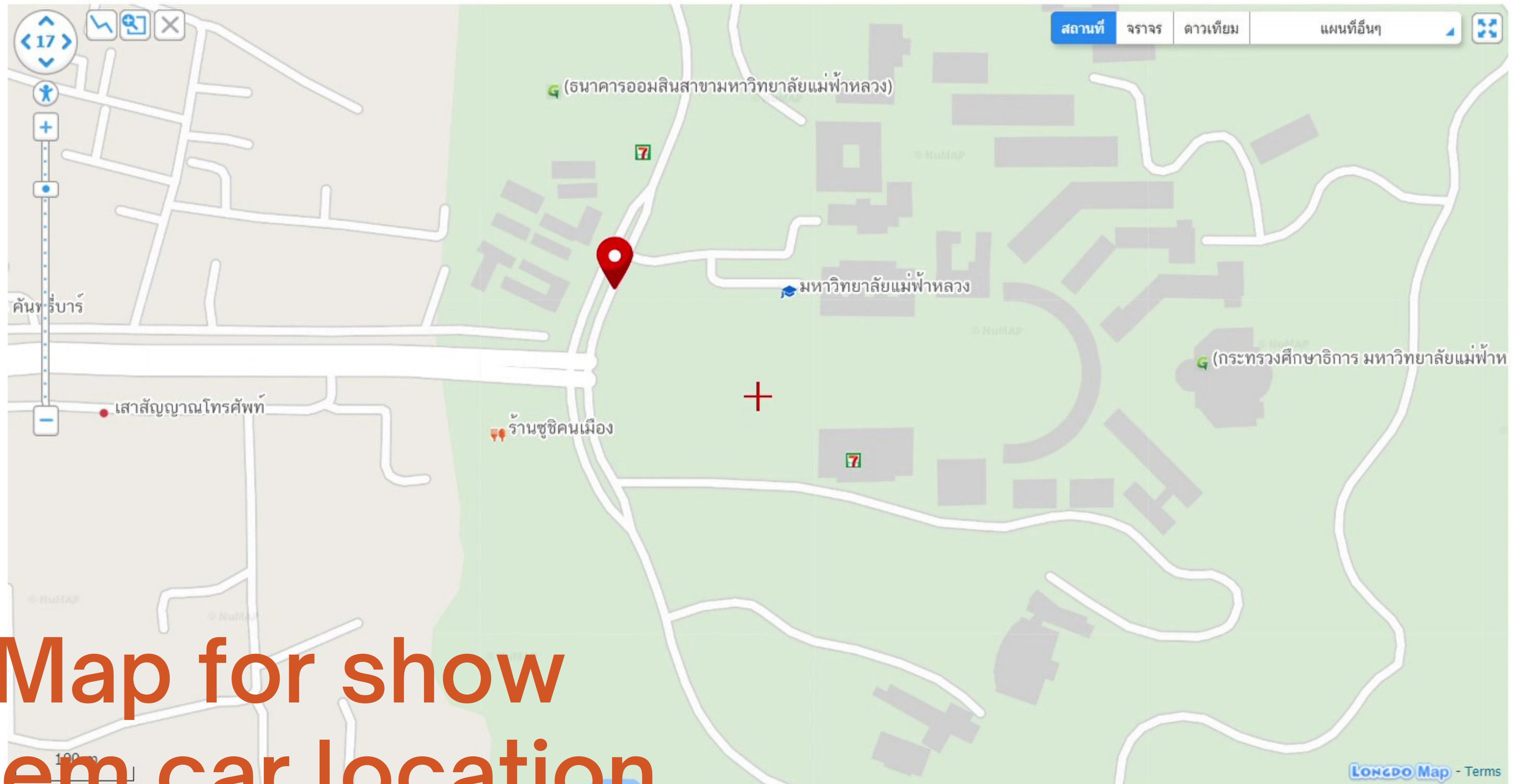


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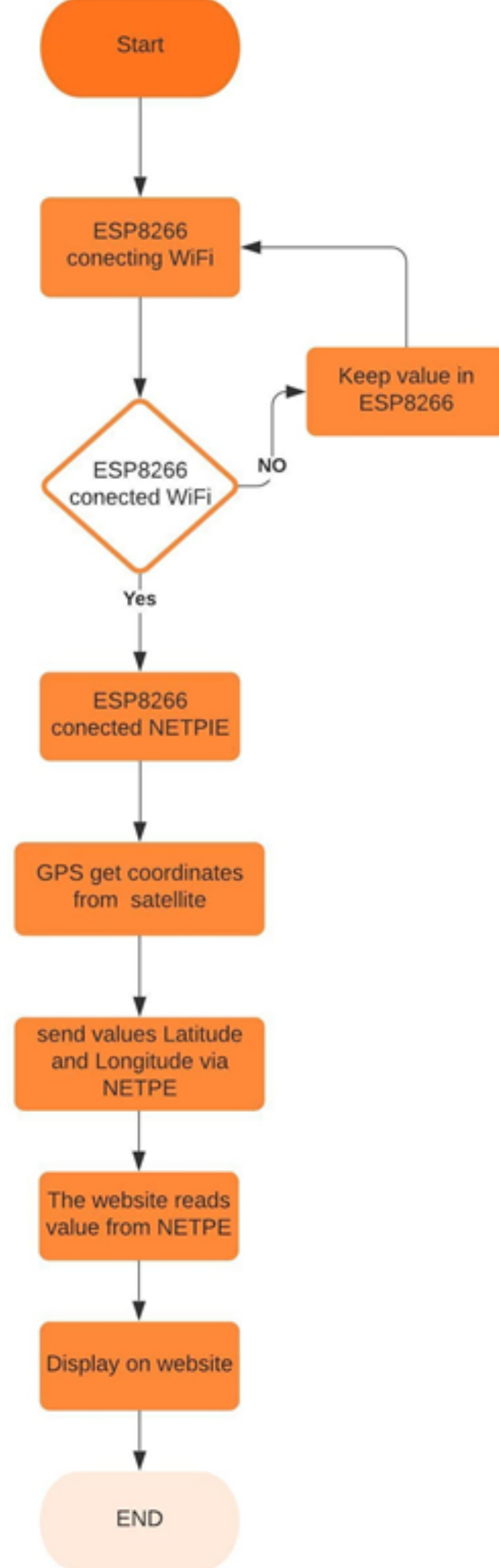


Website design

This website is made for the convenience of using the gem car service because you can check the position of the gem car.



Map for show
gem car location

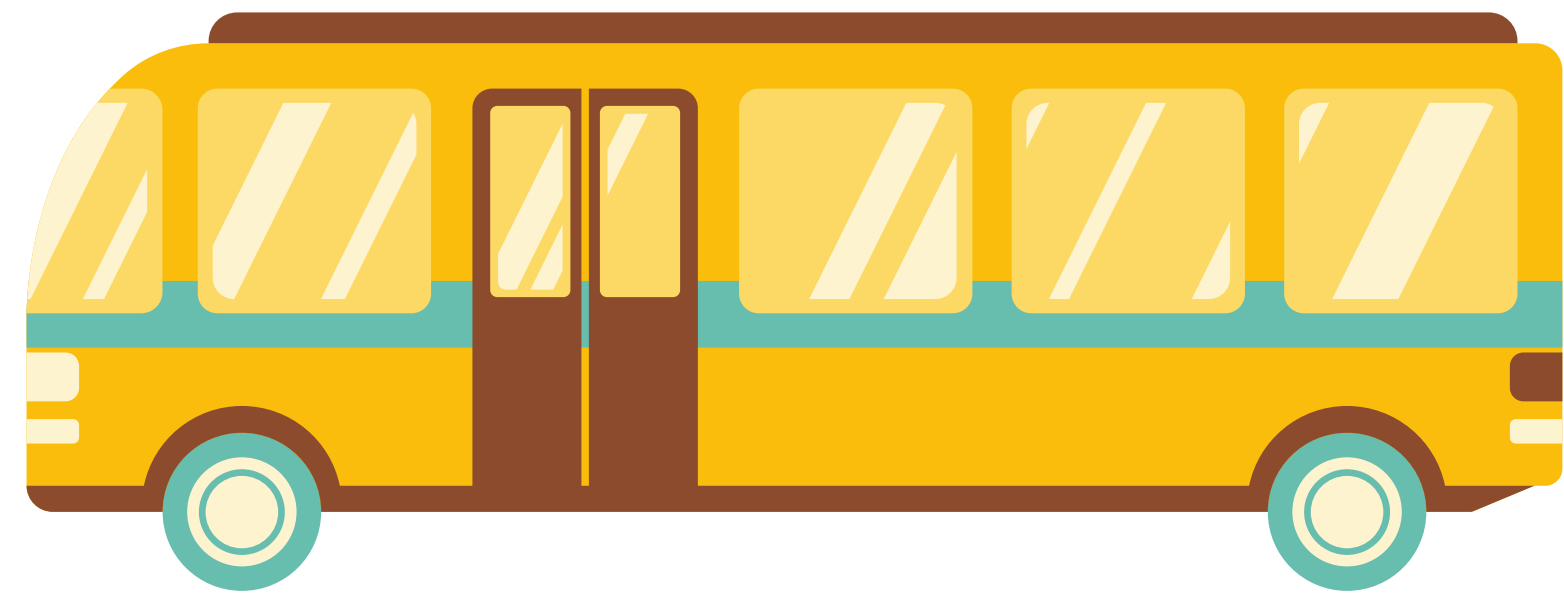


Flowchart



Track and display the status on the website

We use the GPS module to display the position and connect it to the ESP8266 by connecting it to NETPIE (IOT platform) connected to the website.



Project_GPS ▾

Project_GPS / group / all_GPS

 Edit Overview Device List Device Groups Freeboard Event Hooks Setting

Name ▴ ▾

Tags

Create Date ▴ ▾



weblongdoMAP

-

-

2021-02-23 18:20



NodeMCU

-

-

2021-02-12 00:56

1-2 of 2 items



1

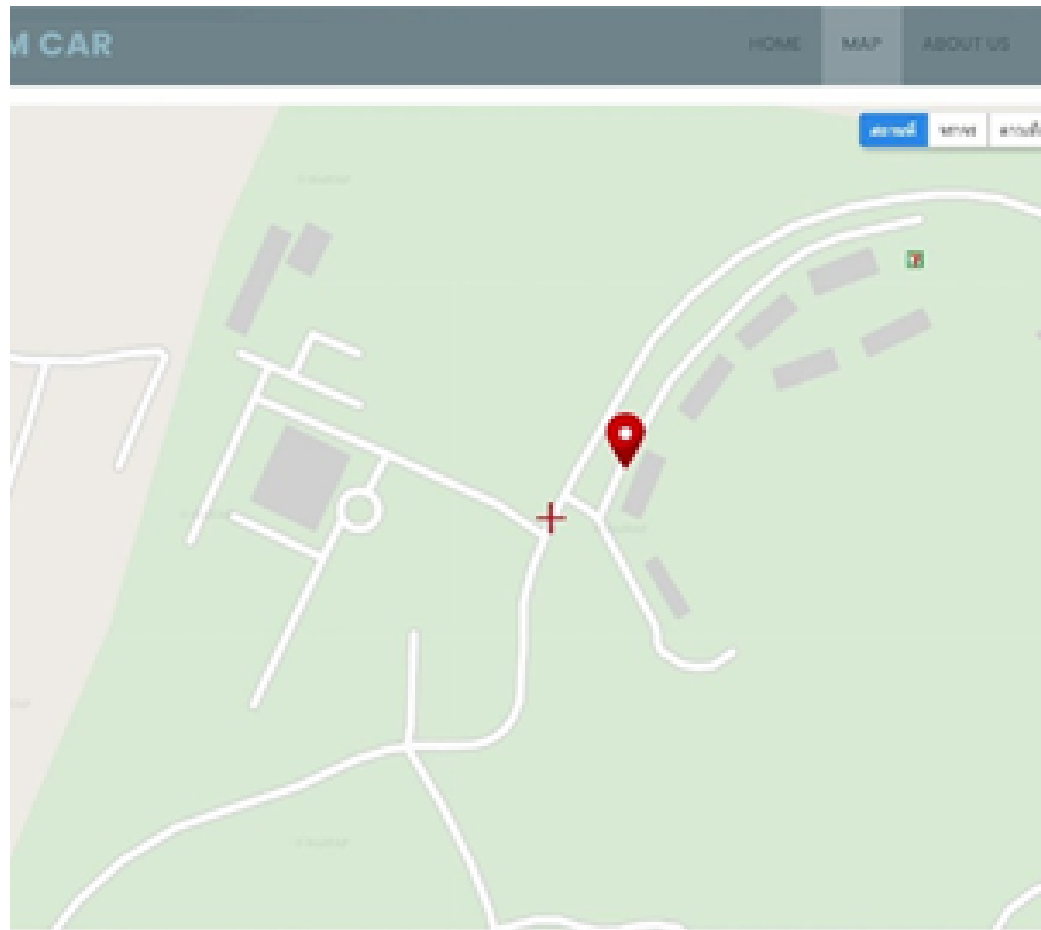


10 / page ▾

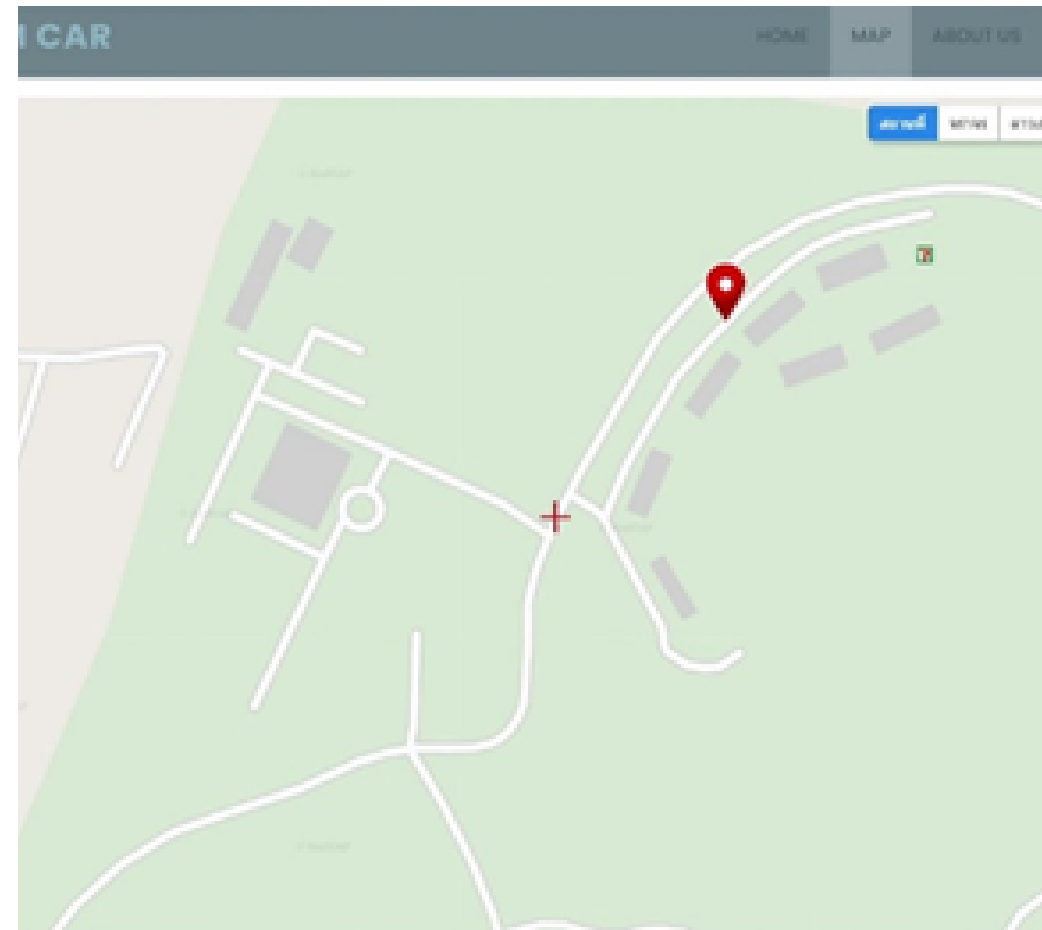
NETPIE2020

Copyright © 2018-2020 Created by NETPIE

To connect ESP8266 and website to NETPIE



Location on the map current
web page



Location on the map the web
page for 5 sec.

**Tested by sending a value from the GPS up
on the web, it will reset on the web every 5
seconds for location.**

Experiment



Experiment Result

Tested by sending a value from the GPS up on the web, it will reset on the web every 5 seconds for location.

Round	Latitude	Longitude	Time
1	20.0567147	99.8969159	14.25
2	20.0569522	99.8971817	14.30
3	20.0573868	99.8974285	14.35
4	20.0580889	99.8979746	14.40

After the test of system, we can know the location detected by GPS on a webpage and resent every 5 seconds on the map

```
23:38:52.189 -> WiFi connected
23:38:52.189 -> IP address:
23:38:52.189 -> 192.168.43.243
23:38:52.189 -> Attempting MQTT connection...connected
23:38:52.423 -> connected
23:38:52.749 -> 20.047329,99.881050
23:38:58.253 -> 20.047333,99.881050
23:39:03.755 -> 20.047331,99.881050
23:39:09.272 -> 20.047331,99.881050
23:39:14.727 -> 20.047333,99.881035
23:39:19.773 -> Attempting MQTT connection...failed try again in 5 seconds
23:39:20.288 -> 20.047333,99.881035
23:39:25.281 -> Attempting MQTT connection...failed try again in 5 seconds
23:39:25.748 -> 20.047335,99.881042
23:39:30.703 -> Attempting MQTT connection...failed try again in 5 seconds
23:39:31.265 -> 20.047337,99.881042
23:39:36.264 -> Attempting MQTT connection...failed try again in 5 seconds
23:39:36.733 -> 20.047337,99.881050
```

After the test to store the LOSS value in the array every 5 seconds so that it can be triggered when the device is disconnected to WIFI.

Thank you for watching the presentation.

PIMNIPA SRIKHAMTHAI
WIRAWAN KHUNMEK
PHAKPHUM SOMSUK

